

Milnor's Exotic 7-Spheres

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Abstract

In this note, we give a detailed exposition of Milnor's construction [4] of exotic smooth structures on the 7-sphere. The key idea is to realize S^7 as the total space of certain S^3 -bundles over S^4 that arise from oriented rank-4 vector bundles classified by elements of $\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$.

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1 Introduction

A fundamental problem arises when studying differentiable manifolds is whether manifolds may admit distinct smooth structures despite sharing the same underlying topological type. The most striking example occurs in dimension 7, where Milnor constructed smooth manifolds that are homeomorphic but not diffeomorphic to the standard sphere S^7 . These manifolds, now known as *Milnor's exotic 7-spheres*, arise as total spaces of certain sphere bundles over S^4 .

The starting point is the classification of oriented rank-4 vector bundles over S^4 . Since

$$\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z},$$

each pair $(i, j) \in \mathbb{Z}^2$ determines a bundle ξ_{ij} via a clutching map $f_{ij} : S^3 \rightarrow SO(4)$ defined by

$$f_{ij}(u)(v) = u^i v u^j, \quad u \in S^3 \subset \mathbb{H}, v \in \mathbb{R}^4 \cong \mathbb{H}.$$

Choosing a metric on ξ_{ij} , we define the 7-manifold $M_{ij} = S(\xi_{ij})$ to be its unit sphere bundle. A natural question is:

For which pairs (i, j) is M_{ij} diffeomorphic to S^7 ?

We show that whenever $i + j = 1$, the manifold M_{ij} is at least homeomorphic to S^7 . This is achieved by constructing a smooth function $f : M_{ij} \rightarrow \mathbb{R}$ with exactly two critical points and invoking Reeb's theorem, which asserts that any closed manifold admitting such a Morse function must be homeomorphic to a sphere.

Determining whether M_{ij} is *diffeomorphic* to S^7 requires finer invariants. Following Milnor, we study the λ -invariant of a closed 7-manifold M , defined using the first Pontrjagin class of a bounding 8-manifold and the signature of its intersection form. We prove that $\lambda(M)$ is independent of the choice of bounding manifold, and we compute $\lambda(M_{ij})$ in terms of the integers (i, j) by first showing

$$p_1(\xi_{ij}) = \pm 2(i - j)[S^4].$$

This, together with the Hirzebruch signature theorem, leads to

$$\lambda(M_{ij}) \equiv (i - j)^2 - 1 \pmod{7}.$$

Since $\lambda(S^7) = 0$, we conclude that whenever $(i - j)^2 \not\equiv 1 \pmod{7}$, the manifold M_{ij} is not diffeomorphic to the standard sphere.

The purpose of this paper is to present a clear, self-contained account of Milnor's construction, emphasizing the roles of clutching functions, characteristic classes, Morse theory, and intersection theory in distinguishing smooth structures on high-dimensional manifolds.

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2 Preliminaries

2.1 Vector Bundles

In this section, we briefly review the basic notions of fiber bundles, vector bundles. These ideas will be used later, especially in the clutching description of oriented 4-bundles over S^4 associated with elements of $\pi_3(SO(4))$.

A **fiber bundle** consists of a quadruple (E, B, π, F) where E is the total space, B is the base space, $\pi : E \rightarrow B$ is a continuous surjection, and F is the typical fiber. For every point $x \in B$, there exists an open neighborhood U and a homeomorphism (local trivialization)

$$\varphi_U : \pi^{-1}(U) \longrightarrow U \times F$$

such that $\pi = \text{pr}_1 \circ \varphi_U$.

On overlaps $U \cap V$, trivializations differ by a **transition function**

$$g_{UV} : U \cap V \longrightarrow \text{Homeo}(F)$$

defined by

$$\varphi_U \circ \varphi_V^{-1}(x, v) = (x, g_{UV}(x)v).$$

These satisfy the **cocycle condition**

$$g_{UV}(x)g_{VW}(x) = g_{UW}(x) \quad \text{on } U \cap V \cap W.$$

A rank n **real (complex) vector bundle** is a fiber bundle whose fiber is $\mathbb{R}^n(\mathbb{C}^n)$ and whose transition functions take values in $GL(n, \mathbb{R})(GL(n, \mathbb{C}))$. A local trivialization is a homeomorphism

$$\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^n (U_\alpha \times \mathbb{C}^n)$$

which is $\mathbb{R}(\mathbb{C})$ -linear on fibers.

A rank k real vector bundle ξ is **oriented** if the transition maps take values in $SO(n) \subset GL^+(n, \mathbb{R})$. In this case, an **orientation** of ξ is an assignment of an orientation for each fiber E_x such that around each $x \in B$, there is a neighborhood U and a local frame $\{s_1, \dots, s_k\}$ such that at each point $x \in U$, $\{s_1(x), \dots, s_k(x)\}$ is the assigned orientation of E_x . Denote by $\bar{\xi}$ the same vector bundle with its orientation reversed.

The following observation is crucial for later uses. Assume $E, \tilde{E}$ are two rank n oriented vector bundles over B with transition functions $\{g_{\alpha\beta}\}, \{\tilde{g}_{\alpha\beta}\}$ respectively. Suppose that there exists a $T \in O(n) \setminus SO(n)$ such that $\tilde{g}_{\alpha\beta} = Tg_{\alpha\beta}T^{-1}$. Assume μ is an orientation of ξ , i.e. at each point $x \in B$, μ_x is a oriented frame of E_x . Then $T\mu$ is an orientation for $\tilde{E}$.

Indeed,

$$\tilde{g}_{\alpha\beta}T\mu = Tg_{\alpha\beta}\mu.$$

Since $T \in O(n) \setminus SO(n)$, then $T\mu$ and μ are opposite. By $g_{\alpha\beta} \in SO(n)$, $Tg_{\alpha\beta}\mu$ and $T\mu$ are the same orientations. Therefore, $T\mu$ is an orientation for $\tilde{E}$. Hence the orientation of $\tilde{E}$ is opposite to that of E .

It is often convenient to construct vector bundles directly from their transition functions. Given an open cover $\{U_\alpha\}$ of B and continuous maps $g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL(n, \mathbb{R})$ satisfying the cocycle condition

$$g_{\alpha\alpha} = id, \quad g_{\alpha\beta} = g_{\beta\alpha}^{-1}, \quad g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma},$$

one constructs a vector bundle by taking the quotient

$$E = \left(\coprod_{\alpha} U_{\alpha} \times \mathbb{R}^n \right) / \sim$$

where $(x, v)_{\beta} \sim (x, g_{\alpha\beta}(x)v)_{\alpha}$. This yields a rank n vector bundle $\pi : E \rightarrow B$ that admits $\{g_{\alpha\beta}\}$ as transition functions.

Let $\pi_E : E \rightarrow B$ and $\pi_F : F \rightarrow B$ be two rank n vector bundles over the same base. A **bundle map** is a continuous map

$$\Phi : E \longrightarrow F$$

such that:

1. $\pi_F \circ \Phi = \pi_E$,
2. for every $x \in B$, the restriction

$$\Phi_x : E_x \longrightarrow F_x$$

is a linear map between the fibers.

A bundle map $\Phi : E \rightarrow F$ is an isomorphism if each $\Phi_x : E_x \rightarrow F_x$ is a linear isomorphism, denoted by $E \cong F$. In this case, the inverse map $\Phi^{-1} : F \rightarrow E$ is also a bundle map. If E, F are oriented vector bundle, a **preserving-orientation isomorphism** between $E \rightarrow F$ if it is a bundle isomorphism that preserves orientation on each fiber. On the other hand, if the isomorphism reverses orientation, we call it a **reversing-orientation isomorphism**.

Two bundles E and F over B are **isomorphic** if there exists a bundle isomorphism between them. Similarly, two oriented manifolds are isomorphic if there exists a preserving-orientation isomorphism between them. Let us denote

$$\text{Vect}_+^k(B) := \{\text{isomorphism classes of rank } k \text{ oriented vector bundle over } B\}.$$

We list several fundamental examples.

Example 1. 1. Trivial bundle of rank k over B is $\epsilon^k \cong B \times \mathbb{R}^k$.

2. Let M be a smooth m -dimensional manifold. For each point $p \in M$, the tangent space $T_p M$ is an m -dimensional vector space. These fit together smoothly to form the tangent bundle

$$TM = \bigsqcup_{p \in M} T_p M, \quad \pi : TM \rightarrow M.$$

3. For $\mathbb{K} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, consider the n -dimensional projective space $\mathbb{K}P^n := (\mathbb{K}^{n+1} \setminus 0)/(v \sim v\lambda)$. We can define the **tautological line bundle** $\gamma_{\mathbb{K}}^n$ over $\mathbb{K}P^n$ as

$$\gamma_{\mathbb{K}}^n := \{([x], v) | [x] \in \mathbb{K}P^n, v \in \text{span}\langle x \rangle\}.$$

A metric on bundle $E \rightarrow B$ is the family of inner product $\langle \cdot, \cdot \rangle_x, x \in B$ defined on each fiber E_x such that $\langle \cdot, \cdot \rangle_x$ depends continuously on $x \in B$.

If a rank n bundle $E \rightarrow B$ has a metric, its associated **sphere bundle** $S(E)$ obtained by restricting each fiber to the unit sphere:

$$S(\xi) = \{v \in E : \|v\| = 1\}.$$

is a fiber bundle with fiber S^{n-1} .

A rank n vector bundle ξ over B is said to be *stably trivial* if there exists some $k \geq 0$ such that

$$\xi \oplus \epsilon^k \cong \epsilon^{n+k},$$

where $\epsilon^k = B \times \mathbb{R}^k$ is the trivial rank- k bundle.

Example 2. The tangent bundle of S^n is stably trivial for all n , since $TS^n \oplus NS^n \cong TS^n \oplus \epsilon^1 = T\mathbb{R}^{n+1}$ is trivial.

2.2 Characteristic classes

Characteristic classes are cohomological invariants associated to vector bundles. They measure how far a vector bundle is from being trivial. In this section, we review the characteristic classes relevant to our later computation of the λ -invariant.

2.2.1 Chern Classes

Given η is a rank k complex vector bundle over B , its **Chern classes** are

$$c_i(\eta) \in H^{2i}(B; \mathbb{Z}), \quad i = 0, \dots, k,$$

with total Chern class

$$c(\eta) = 1 + c_1(\eta) + \dots + c_k(\eta).$$

They are characterized by three axioms:

1. $c(\eta \oplus \theta) = c(\eta)c(\theta)$ (**Whitney sum formula**).
2. $c_i(f^*\eta) = f^*c_i(\eta)$ (**Naturality**).
3. $c_1(\gamma_{\mathbb{C}}^1)$ is the generator of $H^2(\mathbb{C}P^\infty)$. (**Normalization**).

For the explicit construction of Chern classes, please refer to [2] or [7].

2.2.2 Pontrjagin Classes

If $\xi(E \rightarrow B)$ is a real vector bundle, its complexification is $E_{\mathbb{C}} = E \otimes_{\mathbb{R}} \mathbb{C}$. The **Pontrjagin classes** of ξ are defined by

$$p_k(\xi) = (-1)^k c_{2k}(\xi_{\mathbb{C}}) \in H^{4k}(B; \mathbb{Z}).$$

Again, the total Pontrjagin class is

$$p(\xi) = 1 + p_1(\xi) + \dots \in H^*(B, \mathbb{Z}).$$

We list several useful properties of Pontrjagin classes that will be used later.

Proposition 1 ([7]). *Pontrjagin classes satisfy following properties:*

1. **Naturality:**

$$p_k(f^*\xi) = f^*p_k(\xi).$$

2. **Orientation independence:** *Reversing the orientation of ξ does not change $p_k(\xi)$.*

3. **Whitney sum formula:** *$p(\xi \oplus \eta)$ is congruent to $p(\xi)p(\eta)$ modulo elements of order 2, i.e.*

$$2p(\xi \oplus \eta) = 2p(\xi)p(\eta)$$

4. **Stable triviality:** $p_k(\xi \oplus \epsilon^r) = p_k(\xi)$.

As a direct consequence of the naturality, if $\xi \cong \eta$, $p(\xi) = p(\eta)$. By identifying elements with same degrees, Whitney sum formula yields

$$2p_1(\xi \oplus \eta) = 2p_1(\xi) + 2p_1(\eta) \in H^4(B, \mathbb{Z}).$$

If M is a differentiable manifold, then the Pontrjagin classes of M is defined as

$$p_k(M) := p_k(TM).$$

Example 3. Any sphere S^n has stably trivial tangent bundle, so $p_k(TS^n) = 0$ for all $k \geq 1$.

2.3 λ Invariant

This is one of the most sophisticated constructions due to Milnor. From now on, we focus only on a 7-dimensional differentiable manifold M^7 with the property that $H^4(M) = H^3(M) = 0$ (This is true for $M = S^7$). The advantage of having this property is that the connecting map $h : H^4(B, M) \rightarrow H^4(B)$ of the l.e.s.

$$H^3(M) = 0 \longrightarrow H^4(B, M) \xrightarrow{h} H^4(B) \longrightarrow H^4(M) = 0$$

is an isomorphism.

For an orientation $\mu \in H_7(M)$, choose an orientation $\nu \in H_8(B, M)$ such that $\partial\nu = \mu$, where $\partial : H_8(B, M) \rightarrow H_7(M)$ is the connecting homomorphism of l.e.s for the pair (B, M)

As shown in [3], every 7-dimensional closed manifold M^7 is the boundary of some 8-dimensional manifold B^8 . The differentiable structure of a manifold is closely tied to its characteristic classes. The idea comes from the simple observation that if two manifolds are diffeomorphic then two tangent bundles are the same, and hence same characteristic classes. An approach is to look for an appropriate invariant of M^7 that is defined in the bounding manifold B^8 . However, that invariant should be independent of the choice of the manifold B^8 .

Define the intersection form for (B, M) as

$$H^4(B, M)/(\text{torsion}) \times H^4(B, M)/(\text{torsion}) \rightarrow \mathbb{Z}, \quad \alpha \mapsto \langle \nu, \alpha^2 \rangle.$$

Let $\sigma(B)$ be the index of this quadratic form, and we call it the signature of B . Using the isomorphism h above, we can define the following "Pontrjagin number"

$$q(B) = \langle \nu, h^{-1}p_1(B) \rangle.$$

Define $\lambda(M) := q(B) - 2\sigma(M) \pmod{7}$. We are going to show that this is actually an invariant of M .

Theorem 1. *$\lambda(M)$ is independent of the choice of manifold B .*

To prove this theorem, we need the following result.

Theorem 2 (Hirzebruch Signature Theorem). *If N is an 8-dimensional closed manifold, then the signature of C with respect to an orientation $\mu \in H_8(M)$*

$$\sigma(C) = \langle \mu, \frac{1}{45}(7p_2(C) - p_1^2(C)) \rangle.$$

Assume B_1, B_2 are manifolds with boundary M . Fix orientations ν_1, ν_2 for B_1, B_2 respectively. Gluing B_1, B_2 along the boundary M using an orientation-reversing diffeomorphism on M gives a closed manifold C . Choose an orientation ν of C that is consistent with ν_1 of B_1 (and hence consistent with $-\nu_2$).

Applying the Hirzebruch Signature Theorem for C , we get

$$45\sigma(C) = 7\langle \nu, p_2(C) \rangle - \langle \nu, p_1^2(C) \rangle = -q(C) \pmod{7}.$$

Then $\lambda(C) = 2q(C) - \sigma(C) = -7\sigma(C) = 0 \pmod{7}$.

To complete the proof of Theorem 1, we need the following lemma.

Lemma 1. $\lambda(C) = \lambda(B_1) - \lambda(B_2)$.

Proof. The naturality of the Mayer-Vietoris sequence gives the following commutative diagram

$$\begin{array}{ccc}
H^n(B_1, M) \oplus H^n(B_2, M) & \xleftarrow{h \cong} & H^n(C, M) \\
\downarrow h_1 \oplus h_2 & & \downarrow j \\
H^n(B_1) \oplus H^n(B_2) & \xleftarrow{k = (\iota_{B_1}^*) \oplus (-\iota_{B_2}^*)} & H^n(C)
\end{array}$$

For $n = 4$, all of these homomorphisms are isomorphisms. For $\alpha_1 \in H^4(B_1, M), \alpha_2 \in H^4(B_2, M)$, then $\alpha = jh^{-1}(\alpha_1 \oplus \alpha_2) \in H^4(C)$. We have

$$\langle \nu, \alpha^2 \rangle = \langle \nu, jh^{-1}(\alpha_1^2 \oplus \alpha_2^2) \rangle = \langle (h^{-1})_* j_* \nu, \alpha_1^2 \oplus \alpha_2^2 \rangle = \langle \nu_1, \alpha_1^2 \rangle - \langle \nu_2, \alpha_2^2 \rangle. \quad (1)$$

The intersection form of C is the difference between the intersection forms of B_1 and B_2 . Therefore,

$$\sigma(C) = \sigma(B_1) - \sigma(B_2).$$

Set $\alpha_1 = h_1^{-1}(p_1(B_1)) \in H^4(B_1, M), \alpha_2 = h_2^{-1}(p_1(B_2)) \in H^4(B_2, M)$. Since $k = \iota_{B_1}^* - \iota_{B_2}^*$, by the naturality of Pontrjagin classes,

$$k(p_1(C)) = p_1(B_1) \oplus (-p_1(B_2)).$$

The commutativity of the diagram above implies that

$$p_1(C) = jh^{-1}(h_1^{-1}(p_1(B_1)) \oplus h_2^{-1}(-p_1(B_2))) = jh^{-1}(\alpha_1 \oplus (-\alpha_2)).$$

The computation (1) yields

$$q(C) = \langle \nu, p_1^2(C) \rangle = \langle \nu_1, \alpha_1^2 \rangle - \langle \nu_2, \alpha_2^2 \rangle = q(B_1) - q(B_2).$$

This completes the proof. $\square$

Therefore,

$$0 = \lambda(C) = \lambda(B_1) - \lambda(B_2) \Leftrightarrow \lambda(B_1) = \lambda(B_2).$$

This proves Theorem 1.

When $M = S^7$, we can choose $B = D^8$, the 8-dimensional unit ball. Since TD^8 is trivial, every Pontrjagin class vanishes. This follows the vanishing of λ .

Proposition 2. *The λ invariant $\lambda(S^7) = 0$.*

2.4 Basics of Morse theory

Morse theory is a powerful tool for studying the topology of a smooth manifold by examining smooth functions on it, usually thought of as "height functions". As the height varies, the topology of the sublevel sets changes in controlled ways governed by the critical points of the function.

We have the following definitions about non-degenerate critical points.

Definition 1. Given a smooth function $f : M \rightarrow \mathbb{R}$, a point $p \in M$ is called **critical** if $df(p) = 0$.

A critical point p of f is **non-degenerate** if the symmetric bilinear form $\text{Hess}f(p) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p) \right)$ on $T_p M$ is non-degenerate. The **index** of f at p is the index of this bilinear form.

The following lemma of Morse shows that the behavior of f at a non-degenerate critical point can be described by this index.

Theorem 3 (Morse). *If p is a non-degenerate critical point for a function f , there exists a local coordinate chart $(y_1, \dots, y_n)$ centered at p , i.e. $y_i(p) = 0, \forall i$, and in this local coordinate system*

$$f(y) = f(0) - y_1^2 - \dots - y_\lambda^2 + y_{\lambda+1}^2 + \dots + y_n^2,$$

where λ is the index of f at p .

Proof. See Lemma 2.2 in [6] □

Remark 1. In particular, when p is a maximum or minimum point, $\text{Hess}f(p)$ is either positive or negative, that is the index of f at p is either $\lambda = n$ or 0.

Then, near p , the function f is of form

$$f - f(p) = \pm(y_1^2 + \dots + y_n^2).$$

There exists a small ϵ such that $f^{-1}[f(p) - \epsilon, f(p) + \epsilon]$ is a closed n -cell.

Throughout this section, fix a smooth real-valued function f on M . Let

$$M^a := f^{-1}[-\infty, a].$$

The following theorem can be considered to be fundamental in Morse theory.

Theorem 4. *Let $a < b$ and suppose that the set $f^{-1}[a, b]$ contains no critical points of f . Then M^a is diffeomorphic to M^b .*

Proof. See Theorem 3.1 in [6]. □

Combining these two important theorems, we can prove the following theorem due to Reeb.

Theorem 5 (Reeb). *If there exists a Morse function $f : M \rightarrow \mathbb{R}$ on a closed manifold M with only two critical points. Then M is homeomorphic to S^n .*

Proof. Let p, q denote two critical points of f . Since M is closed, two critical points of f must be maximum and minimum points. Assume p, q are minimum and maximum respectively, and without loss of generality, $f(p) = 0$ and $f(q) = 1$. As discussed in Remark 1, there exists an ϵ such that $M^\epsilon = f^{-1}[0, \epsilon]$ and $f^{-1}[1 - \epsilon, 1]$ are two closed n -cells. Since there are no critical points in $f^{-1}[\epsilon, 1 - \epsilon]$, by Theorem 4, $M^{1-\epsilon} \cong M^\epsilon \cong e^n$. Notice that $M = M^{1-\epsilon} \cup f^{-1}[1 - \epsilon, 1]$. Since these two pieces are closed n -cells and agree on the boundary $(f^{-1}(1 - \epsilon))$, M is homeomorphic to S^n . □

2.5 Homotopy groups

This section briefly discusses homotopy theory, which may be relevant in the subsequent sections. For more details, please refer to [1].

Throughout this section, let us assume that X is a path-connected topological space. Recall that two continuous maps $f, g : X \rightarrow Y$ are homotopic if there exists a continuous map $F : [0, 1] \times X \rightarrow Y$ such that $F_0 = f$ and $F_1 = g$. The central object of study in homotopy theory is homotopy groups, of which the fundamental group is a prominent example.

Define

$$\pi_n(X, x_0) := [S^n, X] = \{\text{homotopy classes of maps } (S^n, s_0) \rightarrow (X, x_0)\}.$$

We can equip a group structure on $\pi_n X$ as follows. Let $c : S^n \rightarrow S^n \vee S^n$ be the quotient map obtained by collapsing the equator to a point. For $[f], [g] \in \pi_n(X, x_0)$, For $f, g \in \pi_n(X, x_0)$, define $f \vee g : S^n \vee S^n \rightarrow X$ as the map on $S^n \vee S^n$ that agrees with f on the first summand and agrees with g on the second. Hence, the pullback of $c^*(f \vee g)$ is a map on S^n .

Define

$$[f] + [g] := [c^*(f \vee g)].$$

It can be proved that this is actually a group operation, and moreover, $\pi_n(X, x_0)$ is, up to isomorphism, independent of the base point x_0 . In this case, we can just write $\pi_n X$ and call it the n -th homotopy of X .

Example 4. The homotopy classes of maps $S^n \rightarrow S^n$ can be characterized by its degree, i.e. $f \simeq g : S^n \rightarrow S^n$ iff $\deg(f) = \deg(g)$. Therefore, $\pi_n S^n \cong \mathbb{Z}$.

In particular, when $n = 3$, S^3 can be embedded into the quaternion algebra $\mathbb{H}$. It is obvious that $f_i : S^3 \rightarrow S^3, u \mapsto u^i$ has degree i . Therefore, we have an explicit isomorphism

$$\begin{aligned} \mathbb{Z} &\xrightarrow{\cong} \pi_3 S^3 \\ i &\longmapsto [f_i] \end{aligned}$$

The following basic results can be found [1].

Theorem 6. A covering space projection $p : \tilde{X} \rightarrow X$ induces isomorphism $p_* : \pi_n \tilde{X} \rightarrow \pi_n X, \forall n \geq 2$.

Theorem 7. For an arbitrary collection of path-connected space X_α ,

$$\pi_n(\Pi_\alpha X_\alpha) \cong \Pi_\alpha \pi_n X_\alpha, \forall n.$$

Now, shift the focus to the homotopy groups of $SO(4)$.

Lemma 2. The map $\rho : S^3 \times S^3 \rightarrow SO(4), (a, b) \mapsto (x \mapsto a \cdot x \cdot b)$, where $\cdot$ stands for the quaternion multiplication on $S^3 \subset \mathbb{H}$, is a double cover of $SO(4)$.

Proof. This is proven by using that the kernel of ρ contains two elements $\{(1, 1), (-1, -1)\}$ as in Proposition 3.1.5 [8] and every rotation in $SO(4)$ can be decomposed as $(x \mapsto uxv), u, v \in S^3$ as in [9]. $\square$

Applying Theorems 6, 7, we obtain that

$$\pi_3 SO(4) \cong \pi_3(S^3 \times S^3) \cong \pi_3 S^3 \times \pi_3 S^3 \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Since $SO(4)$ itself possesses a group structure. A natural question to ask is whether there is a connection between the group structures of $SO(4)$ and $\pi_3 SO(4)$. The answer is affirmative. It turns out the group operation in $\pi_3 SO(4)$ is induced by the group operation of $SO(4)$.

Lemma 3. For $[f], [g] \in \pi_3 SO(4)$,

$$[f] + [g] = [f \cdot g]$$

Proof. For each $(i, j) \in \mathbb{Z} \oplus \mathbb{Z}$, define $f_{ij} : S^3 \rightarrow SO(4), u \mapsto (v \mapsto u^i \cdot v \cdot u^j)$. Apparently, by simple computations, $\rho^*([f_i], [f_j]) = [f_{ij}]$, where f_i, f_j are defined in Example 4. As a consequence, $\pi_3 SO(4)$ is generated by $[f_{ij}], \forall i, j$. It is easy to check that $f_{i+i', j+j'} = f_{ij} f_{i'j'}$. Therefore,

$$\begin{aligned} [f_{ij} f_{i'j'}] &= [f_{i+i', j+j'}] = \rho_*([f_{i+i'}], [f_{j+j'}]) = \rho_*([f_i] + [f_{i'}], [f_j] + [f_{j'}]) \\ &= \rho_*([f_i], [f_j]) + \rho_*([f_{i'}], [f_{j'}]) = [f_{ij}] + [f_{i'j'}]. \end{aligned}$$

$\square$

2.6 Clutching functions

In this section, we follow the construction in [2] to produce oriented vector bundles over S^k by maps from $S^{k-1} \rightarrow SO(n)$, which are called **clutching functions**. This construction is useful in Milnor's examples we will discuss later.

Recall that the standard differentiable structure of S^k is obtained by gluing 2 copies of k -dimensional Euclidean spaces $\mathbb{R}_\pm^n$ via the map

$$\begin{aligned} \mathbb{R}_+^k \setminus 0 &\longrightarrow \mathbb{R}_-^k \setminus 0 \\ u &\longmapsto \frac{u}{|u|^2}. \end{aligned}$$

Given a continuous map $f : S^{k-1} \rightarrow SO(n)$, which is called a clutching function, we can construct a vector bundle over S^n such that this bundle is trivialized over $\mathbb{R}_\pm^n$ and the map f defines the transition map.

Let us start with two trivial bundles over $\mathbb{R}_\pm^k$. We glue these two bundles using the clutching function $f : S^{k-1} \rightarrow SO(n)$ as follows. More precisely, the transition function is defined as

$$\begin{aligned} (\mathbb{R}_+^k \setminus 0) \times \mathbb{R}^n &\longrightarrow (\mathbb{R}_-^k \setminus 0) \times \mathbb{R}^n \\ (u, v) &\longmapsto \left(\frac{u}{|u|^2}, f\left(\frac{u}{|u|}\right) v \right). \end{aligned} \tag{2}$$

Denote the resulting bundle by E_f .

Assume that $f, g : S^{k-1} \rightarrow SO(n)$ are two homotopic clutching functions. There exists a homotopy $F : I \times S^{k-1} \rightarrow SO(n)$ such that $F_0 = f$ and $F_1 = g$. By performing a similar construction as above, we can obtain a rank n oriented vector bundle E_F associated with the clutching function F over $I \times S^k$. Let $\iota_t : S^k \hookrightarrow I \times S^k, x \mapsto (t, x)$ be the natural inclusions. Then it is clear that the pullback $\iota_t^* E_F \cong E_{F_t}$. By Proposition 1.7 [2], we obtain $E_f \cong \iota_0^* E_F \cong \iota_1^* E_F \cong E_g$. Thus, E_f is determined uniquely up to isomorphism by the homotopy class $[f] \in [S^{k-1}, SO(n)] = \pi_{k-1} SO(n)$. Moreover, as shown in [2], this construction actually gives us a 1-1 correspondence

$$\pi_{k-1} SO(n) \xleftarrow{\cong} \text{Vect}_+^n(S^k),$$

where $\text{Vect}_+^n(S^k)$ denotes the set of isomorphic classes of rank n oriented vector bundles over S^k .

Example 5. We are going to compute a concrete example obtained from this construction. We will prove that the bundle ξ_{10} constructed by the clutching function $f_{10}(u)v = uv$ is exactly the quaternion tautological line bundle $\gamma_{\mathbb{H}}^1$ over $\mathbb{H}P^1$.

First, we have $\mathbb{H}P^1$ can be decomposed as

$$\mathbb{H}P^1 = \mathbb{H}_+ \cup \mathbb{H}_- \cong \mathbb{R}_+^4 \cup \mathbb{R}_-^4 \cong S^4,$$

where $\mathbb{H}_+ = \{[u : v] | u \neq 0\}$ and $\mathbb{H}_- = \{[u : v] | v \neq 0\}$. Also, the gluing map is

$$\begin{aligned} \mathbb{H}_+ \setminus 0 &\longrightarrow \mathbb{H}_- \setminus 0 \\ [1 : u] &\longmapsto [1/u : 1] \end{aligned}$$

We have two local trivializations over $\mathbb{H}_{\pm}$ as follows.

$$\begin{aligned} \mathbb{R}_+^4 \times \mathbb{R}^4 &\cong \mathbb{H}_+ \times \mathbb{H} \longrightarrow \gamma_{\mathbb{H}}^1|_{\mathbb{H}_+} \\ ([1 : u], v) &\longmapsto ([1 : u], (v, uv)) \\ \mathbb{R}_-^4 \times \mathbb{R}^4 &\cong \mathbb{H}_- \times \mathbb{H} \longrightarrow \gamma_{\mathbb{H}}^1|_{\mathbb{H}_-} \\ ([u' : 1], v') &\longmapsto ([u' : 1], (u'v', v')) \end{aligned}$$

Therefore, the corresponding transition map is

$$\begin{aligned} (\mathbb{H}_+ \setminus 0) \times \mathbb{H} &\longrightarrow \gamma_{\mathbb{H}}^1|_{\mathbb{H}_+ \cap \mathbb{H}_-} \longrightarrow (\mathbb{H}_- \setminus 0) \times \mathbb{H} \\ ([1 : u], v) &\longmapsto ([1 : u], (v, uv)) = ([1/u : 1], (1/u \cdot uv, uv)) \longmapsto ([1/u : 1], uv) \end{aligned}$$

This is exactly the clutching function f_{10} . Thus $\xi_{10} \cong \gamma_{\mathbb{H}}^1$.

3 Milnor's Exotic 7-Spheres

3.1 Construction of examples

Recall from the section 2.5 that

$$\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

For each $(i, j) \in \mathbb{Z} \oplus \mathbb{Z}$, recall the map that has been defined in Lemma 3

$$f_{ij}(u)v = u^i \cdot v \cdot u^j, \quad u \in S^3 \subset \mathbb{H}, v \in \mathbb{R}^4 \equiv \mathbb{H}. \quad (3)$$

Let ξ_{ij} denote the unique rank 4 oriented vector bundle (up to isomorphism) associated with the clutching function $[f_{ij}] \in \pi_3(SO(4))$.

Choose a metric on ξ_{ij} and let $S(\xi_{ij})$ denote the corresponding sphere bundle of ξ_{ij} . Then the total space of $S(\xi_{ij})$ is a 7-dimensional manifold, denoted by M_{ij} .

In the following parts, we are going to prove that for suitable choices of (i, j) , M_{ij} is homeomorphic to S^7 but not diffeomorphic to S^7 .

3.2 M_{ij} is homeomorphic to S^7

The main ingredient to prove M_{ij} is homeomorphic to S^7 is constructing the Morse function so that the Reeb's Theorem (Theorem 5) can be applied.

Recall that S^4 can be decomposed as $S^4 = \mathbb{R}_+^4 \cup \mathbb{R}_-^4$ and that ξ_{ij} is trivialized over $\mathbb{R}^\pm$. By restricting fibers of ξ_{ij} to unit spheres, the clutching function f_{ij} defines the corresponding transition function of $S(\xi_{ij})$ as

$$\begin{aligned} (\mathbb{R}_+^4 \setminus 0) \times S^3 &\longrightarrow (\mathbb{R}_-^4 \setminus 0) \times S^3 \\ (u, v) &\longmapsto (u', v') = \left(\frac{u}{|u|^2}, f_{ij} \left(\frac{u}{|u|} \right) v \right) = \left(\frac{u}{|u|^2}, \frac{u^i \cdot v \cdot u^j}{|u|^{i+j}} \right). \end{aligned}$$

To simplify this expression, we now consider the case **when** $i + j = 1$. Thereby, the transition function becomes $(u', v') = \left(\frac{u}{|u|^2}, u^i \cdot v \cdot u^{-i} \frac{u}{|u|} \right)$.

Inverting v in terms of v' and u' and noting that $\bar{v} = v^{-1}$, we obtain

$$\bar{v} = \frac{u}{|u|} u^{-i} (v')^{-1} u^i = |u| \overline{(u/|u|)^i} u' (v')^{-1} (u/|u|)^i.$$

By using the basic fact for quaternion algebra that $\Re(q \cdot a \cdot \bar{q}) = \Re(a)$ if $|q| = 1$, we get

$$\Re(v) = \Re(\bar{v}) = |u| \Re(u' (v')^{-1}).$$

By changing the coordinates (u', v') to $(u'', v') = (u' (v')^{-1}, v')$, then

$$\Re(u'') = \frac{\Re(v)}{|u|}.$$

Notice that $|u''| = |u'| = |u|^{-1}$. Then we obtain the following equality

$$\frac{\Re(v)}{\sqrt{1 + |u|^2}} = \frac{\Re(u'')}{\sqrt{1 + |u''|^2}}.$$

If we define $f : M_{ij} \rightarrow \mathbb{R}$ such that

$$f(x) = \frac{\Re(v)}{\sqrt{1 + |u|^2}} = \frac{\Re(u'')}{\sqrt{1 + |u''|^2}}.$$

The resulting functions on two patches $\mathbb{R}_\pm^4 \times S^3$ are smooth and agree on the overlap, hence yield a globally smooth function on M_{ij} .

It remains to prove that f is actually a Morse function with exactly two critical points.

Because we have an explicit expression of f on each local patch (u, v) and (u'', v') , we need to find critical points on these two local patches.

For the coordinates (u'', v') , we have $f(u'', v') = \frac{\Re(u'')}{\sqrt{1+|u''|^2}} = \frac{a}{\sqrt{1+a^2+b^2+c^2+d^2}}$, where $u'' = a + bi + bj + ck$.

A simple computation shows that

$$\frac{\partial}{\partial a} f = \frac{1 + b^2 + c^2 + d^2}{(1 + a^2 + b^2 + c^2 + d^2)^{3/2}} > 0.$$

Thus, f has no critical points in the local patch (u'', v') .

Similar, in the coordinates $(u, v) = (a, b, c, d, v)$,

$$f(u, v) = \frac{\Re(v)}{\sqrt{1 + a^2 + b^2 + c^2 + d^2}}.$$

Direct computations imply

$$\frac{\partial}{\partial k} f = -2 \frac{\Re(v)k}{(1 + |u|^2)^{3/2}},$$

where $k = a, b, c, d$.

$$\nabla_v f = \frac{\nabla_v \Re(v)}{\sqrt{1 + |u|^2}}.$$

Therefore, (u, v) is a critical point if $u = 0$ and $\nabla_v \Re(v) = 0$. It is well-known that $\Re : S^3 \rightarrow \mathbb{R}$ has only two critical points $v = \pm 1$.

Therefore, f has only two critical points $(u, v) = (0, \pm 1)$. They are automatically non-degenerate since one is the maximal point, the other corresponds to the minimal point. Applying the Reeb's theorem, we obtain the following theorem.

Theorem 8. *If $i + j = 1$, M_{ij} is homeomorphic to S^7 .*

3.3 M_{ij} is not diffeomorphic to S^7

The basic idea is to try to compute the invariant λ , which is introduced in section 1, of M_{ij} . If the $\lambda(M_{ij}) \neq \lambda(S^7)$, then M_{ij} cannot be diffeomorphic to S^7 .

Recall that

$$\lambda(M^7) = 2q(B^8) - \sigma(B^8) \mod 7,$$

for some 8-manifold B^8 such that $\partial B^8 = M^7$, where $q(B^8)$ is defined as a Pontrjagin number related to the first Pontrjagin class $p_1(B^8)$ only. Hence, the first step is to try to compute this first Pontrjagin class.

Let B_{ij} be the total space of the D^4 -bundle $D(\xi_{ij}) := \{e \in \xi_{ij} \mid \|e\| \leq 1\}$ over S^4 . Then M_{ij} is the boundary of B_{ij} .

Apparently, the inclusion $\iota : B_{ij} \hookrightarrow E_{ij}$ is an embedding, where E_{ij} is the total space of ξ_{ij} . Hence, since E_{ij}, B_{ij} are both differentiable manifolds, it implies

$$TB_{ij} \cong \iota^* TE_{ij}.$$

Moreover, TE_{ij} can always be decomposed as

$$TE_{ij} \cong \pi_{ij}^* \xi_{ij} \oplus \pi_{ij}^* TS^4,$$

where $\pi_{ij} : E_{ij} \rightarrow S^4$ is the natural projection of ξ_{ij} .

Therefore,

$$2p_1(E_{ij}) = 2\pi_{ij}^* p_1(\xi_{ij}) + 2\pi_{ij}^* p_1(S^4). \quad (4)$$

Since S^4 is stably trivial, then $p_1(S^4)$ vanishes. Hence, by the naturality of Pontrjagin classes,

$$2p_1(B_{ij}) = 2\iota^* p_1(E_{ij}) = 2\iota^* \pi_{ij}^* p_1(\xi_{ij}).$$

Therefore, it is crucial to find out what the first Pontrjagin class of ξ_{ij} is. In fact, we can compute it as follows.

Lemma 4. *The first Pontrjagin class of ξ_{ij} is $p_1(\xi_{ij}) \in H^4(S^4) \cong \mathbb{Z}$ is linear in i, j .*

Proof. Recall that we have the following correspondence from the clutching construction

$$\begin{array}{ccc} \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\cong} & \text{Vect}_+^4(S^4) \\ (i, j) & \longmapsto & \xi_{ij} \end{array}$$

By taking the first Pontrjagin class, we obtain the map

$$\begin{array}{ccccc} \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\cong} & \pi_3(SO(4)) & \longrightarrow & H^4(S^4) \\ (i, j) & \longmapsto & f_{ij} & \longmapsto & p_1(\xi_{ij}) \end{array}$$

The conclusion of this proposition is equivalent to that this map is indeed a group homomorphism. There are two things to verify.

First, we prove that the map

$$\begin{array}{ccc} \pi_3(SO(4)) & \longrightarrow & H^4(S^4) \\ [f] & \longmapsto & p_1(\xi_{[f]}) \end{array}$$

is a group homomorphism, where ξ_f is the bundle associated with the homotopy class of the clutching function f . It is equivalent to checking that

$$p_1(\xi_{[f]+[g]}) = p_1(\xi_{[f]}) + p_1(\xi_{[g]}).$$

Consider the map $c : S^4 \rightarrow S^4 \vee S^4$ that crushes an equator to a point. Recall that

$$[f] + [g] = [c^*(f \vee g)].$$

Define a rank 4 oriented vector bundle ξ' over $S^4 \vee S^4$ by clutching along the "equator" $S^3 \vee S^3$ by $f \vee g$. Let $\iota_{1,2} : S^4 \hookrightarrow S^4 \vee S^4$ be the inclusion of the first and second summands S^4 's in $S^4 \vee S^4$, respectively.

It is clear that $\iota_1^* \xi' \cong \xi_{[f]}$ and $\iota_2^* \xi' \cong \xi_{[g]}$. Then $\iota_{1,2}$ induce the isomorphism

$$\begin{aligned} H^4(S^4 \vee S^4) &\xrightarrow{\cong} H^4(S^4) \oplus H^4(S^4) \\ p_1(\xi') &\longmapsto (\iota_1^* p_1(\xi'), \iota_2^* p_1(\xi')) = (p_1(\xi_{[f]}), p_1(\xi_{[g]})). \end{aligned}$$

It is a basic fact that the map c induces

$$\begin{aligned} c^* : H^4(S^4) \oplus H^4(S^4) &\cong H^4(S^4 \vee S^4) \longrightarrow H^4(S^4) \\ (a, b) &\longmapsto a + b. \end{aligned}$$

Hence, by the naturality of Pontrjagin classes,

$$p_1(\xi_{[f]+[g]}) = c^* p_1(\xi') = c^*(p_1(\xi_{[f]}), p_1(\xi_{[g]})) = p_1(\xi_{[f]}) + p_1(\xi_{[g]}).$$

To finish the proof, we still need to prove that $[f_{i+i',j+j'}] = [f_{ij}] + [f_{i'j'}]$. It is immediate from the definition of f_{ij} that

$$f_{i+i',j+j'} = f_{ij} f_{i'j'}.$$

By Lemma 3,

$$[f_{i+i',j+j'}] = [f_{ij} f_{i'j'}] = [f_{ij}] + [f_{i'j'}].$$

□

Lemma 5. *The first Pontrjagin class $p_1(\xi_{ij}) = c(i - j) \in H^4(S^4) \cong \mathbb{Z}$, for some constant c .*

Proof. It is sufficient to prove that ξ_{ij} , $\xi_{-j,-i}$ are isomorphic bundles but with opposite orientations.

If that is the case, since the first Pontrjagin class is independent of orientation, then $p_1(\xi_{ij}) = p_1(\xi_{-j,-i})$.

As shown in Lemma 4, $p_1(\xi_{ij}) = ai + bj$ for $a, b \in \mathbb{Z}$. Then,

$$ai + bj = a(-j) + b(-i), \forall i, j \in \mathbb{Z}.$$

Applying this for $(i, j) = (1, 0), (0, 1)$ yields $a = -b$.

It therefore remains to show that $\xi_{ij}, \xi_{-j,-i}$ are indeed isomorphic via an orientation-reversing bundle isomorphism.

Notice that the clutching function

$$f_{-j,-i}(u)v = u^{-j} \cdot v \cdot u^{-i} = \overline{u^i \cdot \bar{v} \cdot u^j} = (\mu f_{ij}(u)\mu)v = (\mu f_{ij}(u)\mu^{-1})v,$$

where μ is the quaternion conjugation. In the last equality, we used the fact that $\mu^2 = id$.

The clutching functions encode the transition functions of two bundles. Since $f_{ij}, f_{-j, -i}$ are conjugated by an orientation-reversing isomorphism μ , they induce two isomorphic bundles with opposite orientations. This completes the proof. $\square$

Lastly, we need to find the constant c .

Lemma 6. *The constant c in Lemma 5 is ± 2 .*

Proof. Since the constant c is universal for every i, j , we can try to compute $p_1(\xi_{ij})$ for a special (i, j) . Namely, we can pick $i = 1, j = 0$. We have already showed in Example 5 that $\xi_{10} \cong \gamma_{\mathbb{H}}^1$.

We need the following result.

Lemma 7. $p_1(\gamma_{\mathbb{H}}^1) = \pm 2$.

Details of the proof of this lemma can be found in [10].

Since we have $p_1(\xi_{10}) = c$, then

$$c = p_1(\xi_{10}) = p_1(\gamma_{\mathbb{H}}^1) = \pm 2.$$

$\square$

Lemma 8. $\lambda(M_{ij}) = (i - j)^2 - 1 \pmod{7}$.

Proof. Notice that the projection $\pi_{ij} : B_{ij} \rightarrow S^4$ (by an abuse of notation, let π_{ij} still denote the projection of $D(\xi_{ij})$) is a homotopy equivalence. Therefore, it induces an isomorphism

$$\pi_{ij}^* : H^4(S^4) \rightarrow H^4(B_{ij}).$$

Then $\alpha := \pi_{ij}^*[S^4]$ is the generator of $H^4(B_{ij}) \cong \mathbb{Z}$. Now the intersection form is just a 1 by 1 matrix. Therefore, we are free to choose an orientation $\nu \in H^4(B_{ij}, M_{ij})$ so that $\langle \nu, h^{-1}\alpha^2 \rangle = 1$, where $h : H^4(B_{ij}, M_{ij}) \rightarrow H^4(B_{ij})$ is the isomorphism obtained from the cohomology l.e.s., hence $\sigma(B_{ij}) = 1$.

From Lemmas 6 and 5, we have that $p_1(\xi_{ij}) = \pm 2(i - j)[S^4]$. Hence, by (4),

$$2p_1(B_{ij}) = 2\pi_{ij}^*p_1(\xi_{ij}) = \pm 4(i - j)\alpha.$$

This yields

$$q(B_{ij}) = \langle \nu, h^{-1}(4(i - j)^2\alpha^2) \rangle = 4(i - j)^2.$$

Thus,

$$\begin{aligned} \lambda(M_{ij}) &= 2q(B_{ij}) - \sigma(B_{ij}) \pmod{7} \\ &= 8(i - j)^2 - 1 = (i - j)^2 - 1 \pmod{7}. \end{aligned}$$

$\square$

Recall that $\lambda(S^7) = 0$. Then if $(i - j)^2 \not\equiv 1 \pmod{7}$, $\lambda(M_{ij}) \neq \lambda(S^7)$. This proves the following theorem.

Theorem 9. *If $(i - j)^2 \not\equiv 1 \pmod{7}$, then M_{ij} is not diffeomorphic to S^7 with the standard smooth structure.*

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